8 Sampling Methods and the Central Limit Theorem

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THE NIKE annual report says that the average American buys 6.5 pairs of sports shoes per year. Suppose a sample of 81 customers is surveyed and the population standard deviation of sports shoes purchased per year is 2.1 What is the standard error of the mean in this experiment? (See Exercise 45 and LO8-4.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

LO8-1 Explain why populations are sampled and describe four methods to sample a population.

LO8-2 Define sampling error.

LO8-3 Demonstrate the construction of a sampling distribution of the sample mean.

LO8-4 Recite the central limit theorem and define the mean and standard error of the sampling distribution of the sample mean.

LO8-5 Apply the central limit theorem to calculate probabilities.

INTRODUCTION

Chapters 2 through 4 emphasize techniques to describe data. To illustrate these techniques, we organize the profits for the sale of 180 vehicles by the four dealers included in the Applewood Auto Group into a frequency distribution and compute measures of location and dispersion. Such measures as the mean and the standard deviation describe the typical profit and the spread in the profits. In these chapters, the emphasis is on describing the distribution of the data. That is, we describe something that has already happened.

In Chapter 5, we begin to lay the foundation for statistical inference with the study of probability. Recall that in statistical inference our goal is to determine something about a *population* based only on the *sample*. The population is the entire group of individuals or objects under consideration, and the sample is a part or subset of that population. Chapter 6 extends the probability concepts by describing three discrete probability distributions: the binomial, the hypergeometric, and the Poisson. Chapter 7 describes three continuous probability distributions: the uniform, normal, and exponential. Probability distributions encompass all possible outcomes of an experiment and the probability associated with each outcome. We use probability distributions to evaluate the likelihood something occurs in the future.

This chapter begins our study of sampling. Sampling is a process of selecting items from a population so we can use this information to make judgments or inferences about the population. We begin this chapter by discussing methods of selecting a sample from a population. Next, we construct a distribution of the sample mean to understand how the sample means tend to cluster around the population mean. Finally, we show that for any population the shape of this sampling distribution tends to follow the normal probability distribution.

LO8-1
Explain why populations are sampled and describe four methods to sample a population.

SAMPLING METHODS

In Chapter 1, we said the purpose of inferential statistics is to find something about a population based on a sample. A sample is a portion or part of the population of interest. In many cases, sampling is more feasible than studying the entire population. In this section, we discuss the reasons for sampling, and then several methods for selecting a sample.

Reasons to Sample

When studying characteristics of a population, there are many practical reasons why we prefer to select portions or samples of a population to observe and measure. Here are some of the reasons for sampling:

1. **To contact the whole population would be time-consuming.** A candidate for a national office may wish to determine her chances for election. A sample poll using the regular staff and field interviews of a professional polling firm would take only 1 or 2 days. Using the same staff and interviewers and working 7 days a week, it would take nearly 200 years to contact all the voting population! Even if a large staff of interviewers could be assembled, the benefit of contacting all of the voters would probably not be worth the time.
2. **The cost of studying all the items in a population may be prohibitive.** Public opinion polls and consumer testing organizations, such as Harris Interactive Inc., CBS News Polls, and Zogby Analytics, usually contact fewer than 2,000 of the nearly 60 million families in the United States. One consumer panel–type organization charges $40,000 to mail samples and tabulate responses to test a product (such as breakfast cereal, cat food, or perfume). The same product test using all 60 million families would be too expensive to be worthwhile.
3. **The physical impossibility of checking all items in the population.** Some populations are infinite. It would be impossible to check all the water in Lake Erie for bacterial levels, so we select samples at various locations. The populations of fish, birds, snakes, deer, and the like are large and are constantly moving, being

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born, and dying. Instead of even attempting to count all the ducks in Canada or all the fish in Lake Pontchartrain, we make estimates using various techniques—such as counting all the ducks on a pond selected at random, tracking fish catches, or netting fish at predetermined places in the lake.

4. **The destructive nature of some tests.** If the wine tasters at the Sutter Home Winery in California drank all the wine to evaluate the vintage, they would consume the entire crop, and none would be available for sale. In the area of industrial production, steel plates, wires, and similar products must have a certain minimum tensile strength. To ensure that the product meets the minimum standard, the Quality Assurance Department selects a sample from the current production. Each piece is stretched until it breaks and the breaking point (usually measured in pounds per square inch) recorded. Obviously, if all the wire or all the plates were tested for tensile strength, none would be available for sale or use. For the same reason, only a few seeds are tested for germination by Burpee Seeds Inc. prior to the planting season.
5. **The sample results are adequate.** Even if funds were available, it is doubtful the additional accuracy of a 100% sample—that is, studying the entire population—is essential in most problems. For example, the federal government uses a sample of grocery stores scattered throughout the United States to determine the monthly index of food prices. The prices of bread, beans, milk, and other major food items are included in the index. It is unlikely that the inclusion of all grocery stores in the United States would significantly affect the index because the prices of milk, bread, and other major foods usually do not vary by more than a few cents from one chain store to another.

Simple Random Sampling

The most widely used sampling method is a **simple random sampling.**

> **SIMPLE RANDOM SAMPLE** A sample selected so that each item or person in the population has the same chance of being included.

To illustrate the selection process for a simple random sample, suppose the population of interest is the 750 Major League Baseball players on the active rosters of the 30 teams at the end of the 2017 season. The president of the players' union wishes to form a committee of 10 players to study the issue of concussions. One way of ensuring that every player in the population has the same chance of being chosen to serve on the Concussion Committee is to write each name of the 750 players on a slip of paper and place all the slips of paper in a box. After the slips of paper have been thoroughly mixed, the first selection is made by drawing a slip of paper from the box identifying the first player. The slip of paper is not returned to the box. This process is repeated nine more times to form the committee. (Note that the probability of each selection does increase slightly because the slip is not replaced. However, the differences are very small because the population is 750. The probability of each selection is about 0.0013, rounded to four decimal places.)

Of course, the process of writing all the players' names on a slip of paper is very time-consuming. A more convenient method of selecting a random sample is to use a **table of random numbers** such as the one in Appendix B.4. In this case the union president would prepare a list of all 750 players and number each of the players from 1 to 750 with a computer application. Using a table of random numbers, we would randomly pick a starting place in the table, and then select 10 three-digit numbers between 001 and 750. A computer can also generate random numbers. These numbers would correspond with the 10 players in the list that will be asked to participate on the committee. As the name simple random sampling implies, the probability of selecting any number between 001 and 750 is the same. Thus, the probability of selecting the player assigned the number 131 is

STATISTICS IN ACTION

To insure that an unbiased, representative sample is selected from a population, lists of random numbers are needed. In 1927, L. Tippett published the first book of random numbers. In 1938, R. A. Fisher and F. Yates published 15,000 random digits generated using two decks of cards. In 1955, RAND Corporation published a million random digits, generated by the random frequency pulses of an electronic roulette wheel. Since then, computer programs have been developed for generating digits that are "almost" random and hence are called *pseudo-random.* The question of whether a computer program can be used to generate numbers that are truly random remains a debatable issue.

the same as the probability of selecting player 722 or player 382. Using random numbers to select players for the committee removes any bias from the selection process.

The following example shows how to select random numbers using a portion of a random number table illustrated below. First, we choose a starting point in the table. One way of selecting the starting point is to close your eyes and point at a number in the table. Any starting point will do. Another way is to randomly pick a column and row. Suppose the time is 3:04. Using the hour, three o'clock, pick the third column and then, using the minutes, four, move down to the fourth row of numbers. The number is 03759. Because there are only 750 players, we will use the first three digits of a five-digit random number. Thus, 037 is the number of the first player to be a member of the sample. To continue selecting players, we could move in any direction. Suppose we move right. The first three digits of the number to the right of 03759 are 447. Player number 447 is the second player selected to be on the committee. The next three-digit number to the right is 961. You skip 961 as well as the next number 784 because there are only 750 players. The third player selected is number 189. We continue this process until we have 10 players.

50525	57454	28455	68226	34656	38884	39018
72507	53380	53827	42486	54465	71819	91199
34986	74297	00144	38676	89967	98869	39744
68851	27305	03759	44723	96108	78489	18910
06738	62879	03910	17350	49169	03850	18910
11448	10734	05837	24397	10420	16712	94496
		Starting point	Second player			Third player

Statistical packages such as Minitab and spreadsheet packages such as Excel have software that will select a simple random sample. The following example/solution uses Excel to select a random sample from a list of the data.

EXAMPLE

Jane and Joe Miley operate the Foxtrot Inn, a bed and breakfast in Tryon, North Carolina. There are eight rooms available for rent at this B&B. For each day of June 2017, the number of rooms rented is listed. Use Excel to select a sample of five nights during the month of June.

June	Rentals	June	Rentals	June	Rentals
1	0	11	3	21	3
2	2	12	4	22	2
3	3	13	4	23	3
4	2	14	4	24	6
5	3	15	7	25	0
6	4	16	0	26	4
7	2	17	5	27	1
8	3	18	3	28	1
9	4	19	6	29	3
10	7	20	2	30	3

SOLUTION

Excel will select the random sample and report the results. On the first sampled date, four of the eight rooms were rented. On the second sampled date in June, seven rooms were rented. The information is reported in column D of the Excel spreadsheet. The Excel steps are listed in the **Software Commands** in Appendix C. The Excel system performs the sampling *with* replacement. This means it is possible for the same day to appear more than once in a sample.

	A	B	C	D
1	Day of June	Rentals		Sample
2	1	0		4
3	2	2		7
4	3	3		4
5	4	2		3
6	5	3		1
7	6	4		
8	7	2		
9	8	3		
10	9	4		
11	10	7		
12	11	3		
13	12	4		
14	13	4		
15	14	4		

SELF-REVIEW 8–1

The following roster lists the students enrolled in an introductory course in business statistics. Three students will be randomly selected and asked questions about course content and method of instruction.

(a) The numbers 00 through 45 are handwritten on slips of paper and placed in a bowl. The three numbers selected are 31, 7, and 25. Which students are in the sample?

(b) Now use the table of random digits, Appendix B.4, to select your own sample.

(c) What would you do if you encountered the number 59 in the table of random digits?

STAT 264 BUSINESS STATISTICS
9:00 AM - 9:50 AM MW; 118 CARLSON HALL; PROFESSOR LIND

RANDOM NUMBER	NAME	CLASS RANK	RANDOM NUMBER	NAME	CLASS RANK
00	ANDERSON, RAYMOND	SO	23	MEDLEY, CHERYL ANN	SO
01	ANGER, CHERYL RENEE	SO	24	MITCHELL, GREG R	FR
02	BALL, CLAIRE JEANETTE	FR	25	MOLTER, KRISTI MARIE	SO
03	BERRY, CHRISTOPHER G	FR	26	MULCAHY, STEPHEN ROBERT	SO
04	BOBAK, JAMES PATRICK	SO	27	NICHOLAS, ROBERT CHARLES	JR
05	BRIGHT, M. STARR	JR	28	NICKENS, VIRGINIA	SO
06	CHONTOS, PAUL JOSEPH	SO	29	PENNYWITT, SEAN PATRICK	SO
07	DETLEY, BRIAN HANS	JR	30	POTEAU, KRIS E	JR
08	DUDAS, VIOLA	SO	31	PRICE, MARY LYNETTE	SO
09	DULBS, RICHARD ZALFA	JR	32	RISTAS, JAMES	SR
10	EDINGER, SUSAN KEE	SR	33	SAGER, ANNE MARIE	SO
11	FINK, FRANK JAMES	SR	34	SMILLIE, HEATHER MICHELLE	SO
12	FRANCIS, JAMES P	JR	35	SNYDER, LEISHA KAY	SR
13	GAGHEN, PAMELA LYNN	JR	36	STAHL, MARIA TASHERY	SO
14	GOULD, ROBYN KAY	SO	37	ST. JOHN, AMY J	SO
15	GROSENBACHER, SCOTT ALAN	SO	38	STURDEVANT, RICHARD K	SO
16	HEETFIELD, DIANE MARIE	SO	39	SWETYE, LYNN MICHELE	SO
17	KABAT, JAMES DAVID	JR	40	WALASINSKI, MICHAEL	SO
18	KEMP, LISA ADRIANE	FR	41	WALKER, DIANE ELAINE	SO
19	KILLION, MICHELLE A	SO	42	WARNOCK, JENNIFER MARY	SO
20	KOPERSKI, MARY ELLEN	SO	43	WILLIAMS, WENDY A	SO
21	KOPP, BRIDGETTE ANN	SO	44	YAP, HOCK BAN	SO
22	LEHMANN, KRISTINA MARIE	JR	45	YODER, ARLAN JAY	JR

Systematic Random Sampling

The simple random sampling procedure is awkward in some research situations. For example, Stood's Grocery Market needs to sample their customers to study the length of time customers spend in the store. Simple random sampling is not an effective method. Practically, we do not have a list of customers, so assigning random numbers to customers is impossible. Instead, we can use **systematic random sampling** to select a representative sample. Using this method for Stood's Grocery Market, we decide to select 100 customers over 4 days, Monday through Thursday. We will select 25 customers a day and begin the sampling at different times each day: 8 a.m., 11 a.m., 4 p.m., and 7 p.m. We write the 4 times and 4 days on slips of paper and put them in two hats—one hat for the days and the other hat for the times. We select one slip from each hat. This ensures that the time of day is randomly assigned for each day. Suppose we selected 4 p.m. for the starting time on Monday. Next we select a random number between 1 and 10; it is 6. Our selection process begins on Monday at 4 p.m. by selecting the sixth customer to enter the store. Then, we select every 10th (16th, 26th, 36th) customer until we reach the goal of 25 customers. For each of these sampled customers, we measure the length of time the customer spends in the store.

STATISTICS IN ACTION

Random and unbiased sampling methods are extremely important to make valid statistical inferences. In 1936, the *Literary Digest* conducted a straw vote to predict the outcome of the presidential race between Franklin Roosevelt and Alfred Landon. Ten million ballots in the form of returnable postcards were sent to addresses taken from *Literary Digest* subscribers, telephone directories and automobile registrations. In 1936 not many people could afford a telephone or an automobile. Thus, the population that was sampled did not represent the population of voters. A second problem was with the non-responses. More than 10 million people were sent surveys, and more than 2.3 million responded. However, no attempt was made to see whether those responding represented a cross-section of all the voters. On Election Day, Roosevelt won with 61% of the vote. Landon had 39%. In the mid-1930s people who had telephones and drove automobiles clearly did not represent American voters!

SYSTEMATIC RANDOM SAMPLE A random starting point is selected, and then every *k*th member of the population is selected.

Simple random sampling is used in the selection of the days, the times, and the starting point. But the systematic procedure is used to select the actual customer.

Before using systematic random sampling, we should carefully observe the physical order of the population. When the physical order is related to the population characteristic, then systematic random sampling should not be used because the sample could be biased. For example, if we wanted to audit the invoices in a file drawer that were ordered in increasing dollar amounts, systematic random sampling would not guarantee an unbiased random sample. Other sampling methods should be used.

Stratified Random Sampling

When a population can be clearly divided into groups based on some characteristic, we may use **stratified random sampling.** It guarantees each group is represented in the sample. The groups are called **strata.** For example, college students can be grouped as full time or part time; as male or female; or as freshman, sophomore, junior, or senior. Usually the strata are formed based on members' shared attributes or characteristics. A random sample from each stratum is taken in a number proportional to the stratum's size when compared to the population. Once the strata are defined, we apply simple random sampling within each group or stratum to collect the sample.

STRATIFIED RANDOM SAMPLE A population is divided into subgroups, called strata, and a sample is randomly selected from each stratum.

For instance, we might study the advertising expenditures for the 352 largest companies in the United States. The objective of the study is to determine whether firms with high returns on equity (a measure of profitability) spend more on advertising than firms with low returns on equity. To make sure the sample is a fair representation of the 352 companies, the companies are grouped on percent return on equity. Table 8–1 shows the strata and the relative frequencies. If simple random sampling is used, observe that firms in the 3rd and 4th strata have a high chance of selection (probability of 0.87) while firms in the other strata have a small chance of selection (probability of 0.13). We might not select any firms in stratum 1 or 5 *simply by chance*. However, stratified random sampling will guarantee that at least one firm in each of strata 1 and

TABLE 8–1 Number Selected for a Proportional Stratified Random Sample

Stratum	Profitability (return on equity)	Number of Firms	Relative Frequency	Number Sampled
1	30% and over	8	0.02	1*
2	20 up to 30%	35	0.10	5*
3	10 up to 20%	189	0.54	27
4	0 up to 10%	115	0.33	16
5	Deficit	5	0.01	1
Total		352	1.00	50

*0.02 of 50 = 1, 0.10 of 50 = 5, etc.

5 is represented in the sample. Let's say that 50 firms are selected for intensive study. Then based on probability, 1 firm, or (0.02)(50), should be randomly selected from stratum 1. We would randomly select 5, or (0.10)(50), firms from stratum 2. In this case, the number of firms sampled from each stratum is proportional to the stratum's relative frequency in the population. Stratified sampling has the advantage, in some cases, of more accurately reflecting the characteristics of the population than does simple random or systematic random sampling.

Cluster Sampling

Another common type of sampling is **cluster sampling.** It is often employed to reduce the cost of sampling a population scattered over a large geographic area.

CLUSTER SAMPLING A population is divided into clusters using naturally occurring geographic or other boundaries. Then, clusters are randomly selected and a sample is collected by randomly selecting from each cluster.

Suppose you want to determine the views of residents in the greater Chicago, Illinois, metropolitan area about state and federal environmental protection policies. Selecting a random sample of residents in this region and personally contacting each one would be time-consuming and very expensive. Instead, you could employ cluster sampling by subdividing the region into small units, perhaps by counties. These are often called *primary units.*

There are 12 counties in the greater Chicago metropolitan area. Suppose you randomly select 3 counties. The 3 chosen are La Porte, Cook, and Kenosha (see Chart 8–1 below). Next, you select a random sample of the residents in each of these counties and interview them. This is also referred to as sampling through an *intermediate unit.* In this case, the intermediate unit is the county. (Note that this is a combination of cluster sampling and simple random sampling.)

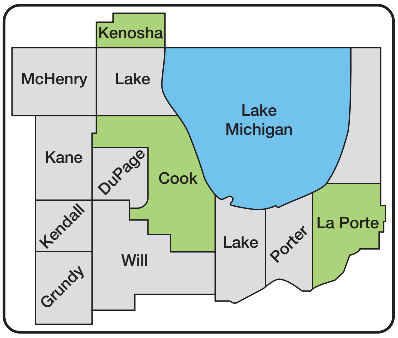


CHART 8–1 The Counties of the Greater Chicago, Illinois, Metropolitan Area

The discussion of sampling methods in the preceding sections did not include all the sampling methods available to a researcher. Should you become involved in a major research project in marketing, finance, accounting, or other areas, you would need to consult books devoted solely to sample theory and sample design.

SELF-REVIEW 8–2

Refer to Self-Review 8–1 and the class roster on page 254. Suppose a systematic random sample will select every ninth student enrolled in the class. Initially, the fourth student on the list was selected at random. That student is numbered 03. Remembering that the random numbers start with 00, which students will be chosen to be members of the sample?

EXERCISES

1. The following is a list of 24 Marco's Pizza stores in Lucas County. The stores are identified by numbering them 00 through 23. Also noted is whether the store is corporate-owned (C) or manager-owned (M). A sample of four locations is to be selected and inspected for customer convenience, safety, cleanliness, and other features.

ID No.	Address	Type	ID No.	Address	Type
00	2607 Starr Av	C	12	2040 Ottawa River Rd	C
01	309 W Alexis Rd	C	13	2116 N Reynolds Rd	C
02	2652 W Central Av	C	14	3678 Rugby Dr	C
03	630 Dixie Hwy	M	15	1419 South Av	C
04	3510 Dorr St	C	16	1234 W Sylvania Av	C
05	5055 Glendale Av	C	17	4624 Woodville Rd	M
06	3382 Lagrange St	M	18	5155 S Main	M
07	2525 W Laskey Rd	C	19	106 E Airport Hwy	C
08	303 Louisiana Av	C	20	6725 W Central	M
09	149 Main St	C	21	4252 Monroe	C
10	835 S McCord Rd	M	22	2036 Woodville Rd	C
11	3501 Monroe St	M	23	1316 Michigan Av	M

a. The random numbers selected are 08, 18, 11, 54, 02, 41, and 54. Which stores are selected?
b. Use the table of random numbers to select your own sample of locations.
c. Using systematic random sampling, every seventh location is selected starting with the third store in the list. Which locations will be included in the sample?
d. Using stratified random sampling, select three locations. Two should be corporate-owned and one should be manager-owned.

2. The following is a list of 29 hospitals in the Cincinnati, Ohio, and Northern Kentucky region. Each hospital is assigned a number, 00 through 28. The hospitals are classified by type, either a general medical/surgical hospital (M/S) or a specialty hospital (S). We are interested in estimating the average number of full- and part-time nurses employed in the area hospitals.

ID Number	Name	Address	Type	ID Number	Name	Address	Type
00	Bethesda North	10500 Montgomery Cincinnati, Ohio 45242	M/S	04	Mercy Hospital– Hamilton	100 Riverfront Plaza Hamilton, Ohio 45011	M/S
01	Ft. Hamilton–Hughes	630 Eaton Avenue Hamilton, Ohio 45013	M/S	05	Middletown Regional	105 McKnight Drive Middletown, Ohio 45044	M/S
02	Jewish Hospital– Kenwood	4700 East Galbraith Rd. Cincinnati, Ohio 45236	M/S	06	Clermont Mercy Hospital	3000 Hospital Drive Batavia, Ohio 45103	M/S
03	Mercy Hospital– Fairfield	3000 Mack Road Fairfield, Ohio 45014	M/S	07	Mercy Hospital– Anderson	7500 State Road Cincinnati, Ohio 45255	M/S

ID Number	Name	Address	Type
08	Bethesda Oak Hospital	619 Oak Street Cincinnati, Ohio 45206	M/S
09	Children's Hospital Medical Center	3333 Burnet Avenue Cincinnati, Ohio 45229	M/S
10	Christ Hospital	2139 Auburn Avenue Cincinnati, Ohio 45219	M/S
11	Deaconess Hospital	311 Straight Street Cincinnati, Ohio 45219	M/S
12	Good Samaritan Hospital	375 Dixmyth Avenue Cincinnati, Ohio 45220	M/S
13	Jewish Hospital	3200 Burnet Avenue Cincinnati, Ohio 45229	M/S
14	University Hospital	234 Goodman Street Cincinnati, Ohio 45267	M/S
15	Providence Hospital	2446 Kipling Avenue Cincinnati, Ohio 45239	M/S
16	St. Francis–St. George Hospital	3131 Queen City Avenue Cincinnati, Ohio 45238	M/S
17	St. Elizabeth Medical Center, North Unit	401 E. 20th Street Covington, Kentucky 41014	M/S
18	St. Elizabeth Medical Center, South Unit	One Medical Village Edgewood, Kentucky 41017	M/S
19	St. Luke's Hospital West	7380 Turfway Drive Florence, Kentucky 41075	M/S
20	St. Luke's Hospital East	85 North Grand Avenue Ft. Thomas, Kentucky 41042	M/S
21	Care Unit Hospital	3156 Glenmore Avenue Cincinnati, Ohio 45211	S
22	Emerson Behavioral Science	2446 Kipling Avenue Cincinnati, Ohio 45239	S
23	Pauline Warfield Lewis Center for Psychiatric Treat.	1101 Summit Road Cincinnati, Ohio 45237	S
24	Children's Psychiatric No. Kentucky	502 Farrell Drive Covington, Kentucky 41011	S
25	Drake Center Rehab—Long Term	151 W. Galbraith Road Cincinnati, Ohio 45216	S
26	No. Kentucky Rehab Hospital—Short Term	201 Medical Village Edgewood, Kentucky	S
27	Shriners Burns Institute	3229 Burnet Avenue Cincinnati, Ohio 45229	S
28	VA Medical Center	3200 Vine Cincinnati, Ohio 45220	S

a. A sample of five hospitals is to be randomly selected. The random numbers are 09, 16, 00, 49, 54, 12, and 04. Which hospitals are included in the sample?
b. Use a table of random numbers to develop your own sample of five hospitals.
c. Using systematic random sampling, every fifth location is selected starting with the second hospital in the list. Which hospitals will be included in the sample?
d. Using stratified random sampling, select five hospitals. Four should be medical and surgical hospitals and one should be a specialty hospital. Select an appropriate sample.

3. Listed below are the 35 members of the Metro Toledo Automobile Dealers Association. We would like to estimate the mean revenue from dealer service departments. The members are identified by numbering them 00 through 34.

ID Number	Dealer
00	Dave White Acura
01	Autofair Nissan
02	Autofair Toyota-Suzuki
03	George Ball's Buick GMC Truck
04	York Automotive Group
05	Bob Schmidt Chevrolet
06	Bowling Green Lincoln Mercury Jeep Eagle
07	Brondes Ford
08	Brown Honda
09	Brown Mazda
10	Charlie's Dodge
11	Thayer Chevrolet/Toyota
12	Spurgeon Chevrolet Motor Sales, Inc.
13	Dunn Chevrolet
14	Don Scott Chevrolet
15	Dave White Chevrolet Co.
16	Dick Wilson Infinity
17	Doyle Buick
18	Franklin Park Lincoln Mercury
19	Genoa Motors
20	Great Lakes Ford Nissan
21	Grogan Towne Chrysler
22	Hatfield Motor Sales
23	Kistler Ford, Inc.
24	Lexus of Toledo
25	Mathews Ford Oregon, Inc.
26	Northtown Chevrolet
27	Quality Ford Sales, Inc.
28	Rouen Chrysler Jeep Eagle
29	Saturn of Toledo
30	Ed Schmidt Jeep Eagle
31	Southside Lincoln Mercury
32	Valiton Chrysler
33	Vin Divers
34	Whitman Ford

a. We want to select a random sample of five dealers. The random numbers are 05, 20, 59, 21, 31, 28, 49, 38, 66, 08, 29, and 02. Which dealers would be included in the sample?

b. Use the table of random numbers to select your own sample of five dealers.
c. Using systematic random sampling, every seventh dealer is selected starting with the fourth dealer in the list. Which dealers are included in the sample?

4. Listed next are the 27 Nationwide Insurance agents in the El Paso, Texas metropolitan area. The agents are numbered 00 through 26. We would like to estimate the mean number of years employed with Nationwide.

ID Number	Agent	ID Number	Agent	ID Number	Agent
00	**Bly Scott** 3332 W Laskey Rd	10	**Heini Bernie** 7110 W Centra	20	**Schwab Dave** 572 W Dussel Dr
01	**Coyle Mike** 5432 W Central Av	11	**Hinckley Dave** 14 N Holland Sylvania Rd	21	**Seibert John H** 201 S Main
02	**Denker Brett** 7445 Airport Hwy	12	**Joehlin Bob** 3358 Navarre Av	22	**Smithers Bob** 229 Superior St
03	**Denker Rollie** 7445 Airport Hwy	13	**Keisser David** 3030 W Sylvania Av	23	**Smithers Jerry** 229 Superior St
04	**Farley Ron** 1837 W Alexis Rd	14	**Keisser Keith** 5902 Sylvania Av	24	**Wright Steve** 105 S Third St
05	**George Mark** 7247 W Central Av	15	**Lawrence Grant** 342 W Dussel Dr	25	**Wood Tom** 112 Louisiana Av
06	**Gibellato Carlo** 6616 Monroe St	16	**Miller Ken** 2427 Woodville Rd	26	**Yoder Scott** 6 Willoughby Av
07	**Glemser Cathy** 5602 Woodville Rd	17	**O'Donnell Jim** 7247 W Central Av		
08	**Green Mike** 4149 Holland Sylvania Rd	18	**Priest Harvey** 5113 N Summit St		
09	**Harris Ev** 2026 Albon Rd	19	**Riker Craig** 2621 N Reynolds Rd		

a. We want to select a random sample of four agents. The random numbers are 02, 59, 51, 25, 14, 29, 77, 69, and 18. Which dealers would be included in the sample?
b. Use the table of random numbers to select your own sample of four agents.
c. Using systematic random sampling, every fifth dealer is selected starting with the third dealer in the list. Which dealers are included in the sample?

LO8-2
Define sampling error.

SAMPLING "ERROR"

In the previous section, we discussed sampling methods that are used to select a sample that is an unbiased representation of the population. In each method, the selection of every possible sample of a specified size from a population has a known chance or probability. This is another way to describe an unbiased sampling method.

Samples are used to estimate population characteristics. For example, the mean of a sample is used to estimate the population mean. However, since the sample is a part or portion of the population, it is unlikely that the sample mean would be *exactly equal* to the population mean. Similarly, it is unlikely that the sample standard deviation would be *exactly equal* to the population standard deviation. We can therefore expect a difference between a *sample statistic* and its corresponding *population parameter*. This difference is called **sampling error.**

SAMPLING ERROR The difference between a sample statistic and its corresponding population parameter.

The following example/solution clarifies the idea of sampling error.

EXAMPLE

Refer to the example/solution on page 253, where we studied the number of rooms rented at the Foxtrot Inn bed and breakfast in Tryon, North Carolina. The population is the number of rooms rented each of the 30 days in June 2017. Find the mean of the population. Select three random samples of 5 days. Calculate the mean rooms rented for each sample and compare it to the population mean. What is the sampling error in each case?

SOLUTION

During the month, there were a total of 94 rentals. So the mean number of units rented per night is 3.13. This is the population mean. Hence we designate this value with the Greek letter μ.

$$\mu = \frac{\Sigma x}{N} = \frac{0 + 2 + 3 + \cdots + 3}{30} = \frac{94}{30} = 3.13$$

The first random sample of five nights resulted in the following number of rooms rented: 4, 7, 4, 3, and 1. The mean of this sample is 3.80 rooms, which we designate as $\bar{x}_1$. The bar over the *x* reminds us that it is a sample mean and the subscript 1 indicates it is the mean of the first sample.

$$\bar{x}_1 = \frac{\Sigma x}{n} = \frac{4 + 7 + 4 + 3 + 1}{5} = \frac{19}{5} = 3.80$$

The sampling error for the first sample is the difference between the population mean (3.13) and the first sample mean (3.80). Hence, the sampling error is $(\bar{x}_1 - \mu) = 3.80 - 3.13 = 0.67$. The second random sample of 5 days from the population of all 30 days in June revealed the following number of rooms rented: 3, 3, 2, 3, and 6. The mean of these five values is 3.40, found by

$$\bar{x}_2 = \frac{\Sigma x}{n} = \frac{3 + 3 + 2 + 3 + 6}{5} = 3.40$$

The sampling error is $(\bar{x}_2 - \mu) = 3.4 - 3.13 = 0.27$. In the third random sample, the mean was 1.80 and the sampling error was -1.33.

Each of these differences, 0.67, 0.27, and -1.33, is the sampling error made in estimating the population mean. Sometimes these errors are positive values, indicating that the sample mean overestimated the population mean; other times they are negative values, indicating the sample mean was less than the population mean.

June	Rentals	June	Rentals	June	Rentals		Sample 1	Sample 2	Sample 3
1	0	11	3	21	3		4	3	0
2	2	12	4	22	2		7	3	0
3	3	13	4	23	3		4	2	3
4	2	14	4	24	6		3	3	3
5	3	15	7	25	0		1	6	3
6	4	16	0	26	4	Total	19	17	9
7	2	17	5	27	1	Mean	3.80	3.40	1.80
8	3	18	3	28	1	Sampling Error	0.67	0.27	-1.33
9	4	19	6	29	3				
10	7	20	2	30	3				

In this case, where we have a population of 30 values and samples of 5 values, there is a very large number of possible samples—142,506 to be exact! To find this value, use the combination formula (5–10) on page 164. Each of the 142,506 different samples has the same chance of being selected. Each sample may have a different sample mean and therefore a different sampling error. The value of the sampling error is based on the particular one of the 142,506 different possible samples selected. Therefore, the sampling errors are random and occur by chance. If you summed the sampling errors for all 142,506 samples, the result would equal zero. This is true because the sample mean is an *unbiased estimator* of the population mean.

LO8-3
Demonstrate the construction of a sampling distribution of the sample mean.

SAMPLING DISTRIBUTION OF THE SAMPLE MEAN

In the previous section, we defined sampling error and presented the results when we compared a sample statistic, such as the sample mean, to the population mean. To put it another way, when we use the sample mean to estimate the population mean, how can we determine how accurate the estimate is? How does:

- A quality-assurance supervisor decide if a machine is filling 20-ounce bottles with 20 ounces of cola based only on a sample of 10 filled bottles?
- **FiveThirtyEight.com** or **Gallup** make accurate statements about the demographics of voters in a presidential race based on relatively small samples from a voting population of nearly 90 million?

To answer these questions, we first develop a *sampling distribution of the sample mean*.

The sample means in the previous example/solution varied from one sample to the next. The mean of the first sample of 5 days was 3.80 rooms, and the second sample mean was 3.40 rooms. The population mean was 3.13 rooms. If we organized the means of all possible samples of 5 days into a probability distribution, the result is called the **sampling distribution of the sample mean.**

SAMPLING DISTRIBUTION OF THE SAMPLE MEAN A probability distribution of all possible sample means of a given sample size.

The following example/solution illustrates the construction of a sampling distribution of the sample mean. We have intentionally used a small population to highlight the relationship between the population mean and the various sample means.

EXAMPLE

Tartus Industries has seven production employees (considered the population). The hourly earnings of each employee are given in Table 8–2.

TABLE 8–2 Hourly Earnings of the Production Employees of Tartus Industries

Employee	Hourly Earnings	Employee	Hourly Earnings
Joe	$14	Jan	14
Sam	14	Art	16
Sue	16	Ted	18
Bob	16		

1. What is the population mean?
2. What is the sampling distribution of the sample mean for samples of size 2?
3. What is the mean of the sampling distribution?
4. What observations can be made about the population and the sampling distribution?

SOLUTION

Here are the solutions to the questions.

1. The population is small so it is easy to calculate the population mean. It is $15.43, found by:

$$\mu = \frac{\Sigma x}{N} = \frac{\$14 + \$14 + \$16 + \$16 + \$14 + \$16 + \$18}{7} = \$15.43$$

We identify the population mean with the Greek letter μ. Recall from earlier chapters, Greek letters are used to represent population parameters.

2. To arrive at the sampling distribution of the sample mean, we need to select all possible samples of 2 without replacement from the population, then compute the mean of each sample. There are 21 possible samples, found by using formula (5–10) on page 164.

$$_{N}C_{n} = \frac{N!}{n!(N-n)!} = \frac{7!}{2!(7-2)!} = 21$$

where $N = 7$ is the number of items in the population and $n = 2$ is the number of items in the sample.

TABLE 8–3 Sample Means for All Possible Samples of 2 Employees

Sample	Employees	Hourly Earnings	Sum	Mean	Sample	Employees	Hourly Earnings	Sum	Mean
1	Joe, Sam	\$14, \$14	\$28	\$14	12	Sue, Bob	16,16	32	16
2	Joe, Sue	14, 16	30	15	13	Sue, Jan	16,14	30	15
3	Joe, Bob	14, 16	30	15	14	Sue, Art	16,16	32	16
4	Joe, Jan	14, 14	28	14	15	Sue, Ted	16,18	34	17
5	Joe, Art	14, 16	30	15	16	Bob, Jan	16,14	30	15
6	Joe, Ted	14, 18	32	16	17	Bob, Art	16,16	32	16
7	Sam, Sue	14, 16	30	15	18	Bob, Ted	16,18	34	17
8	Sam, Bob	14, 16	30	15	19	Jan, Art	14,16	30	15
9	Sam, Jan	14,14	28	14	20	Jan, Ted	14,18	32	16
10	Sam, Art	14,16	30	15	21	Art, Ted	16,18	34	17
11	Sam, Ted	14,18	32	16					

The 21 sample means from all possible samples of 2 that can be drawn from the population of 7 employees are shown in Table 8–3. These 21 sample means are used to construct a probability distribution. This is called the sampling distribution of the sample mean, and it is summarized in Table 8–4.

TABLE 8–4 Sampling Distribution of the Sample Mean for $n = 2$

Sample Mean	Number of Means	Probability
\$14	3	.1429
15	9	.4285
16	6	.2857
17	3	.1429
	21	1.0000

3. Using the data in Table 8–3, the mean of the sampling distribution of the sample mean is obtained by summing the various sample means and dividing the sum by the number of samples. The mean of all the sample means is usually written $\mu_{\bar{x}}$. The μ reminds us that it is a population value because we have considered all possible samples of two employees from the population of seven employees. The subscript $\bar{x}$ indicates that it is the sampling distribution of the sample mean.

$$\mu_{\bar{x}} = \frac{\text{Sum of all sample means}}{\text{Total number of samples}} = \frac{\$14 + \$15 + \$15 + \cdots + \$16 + \$17}{21}$$

$$= \frac{\$324}{21} = \$15.43$$

4. Refer to Chart 8–2. It shows the population distribution based on the data in Table 8–2 and the distribution of the sample mean based on the data in Table 8–4. These observations can be made:
   a. The mean of the distribution of the sample mean ($15.43) is equal to the mean of the population: $\mu = \mu_{\bar{x}}$.
   b. The spread in the distribution of the sample mean is less than the spread in the population values. The sample means range from $14 to $17 while the population values vary from $14 up to $18. If we continue to increase the sample size, the spread of the distribution of the sample mean becomes smaller.
   c. The shape of the sampling distribution of the sample mean and the shape of the frequency distribution of the population values are different. The distribution of the sample mean tends to be more bell-shaped and to approximate the normal probability distribution.

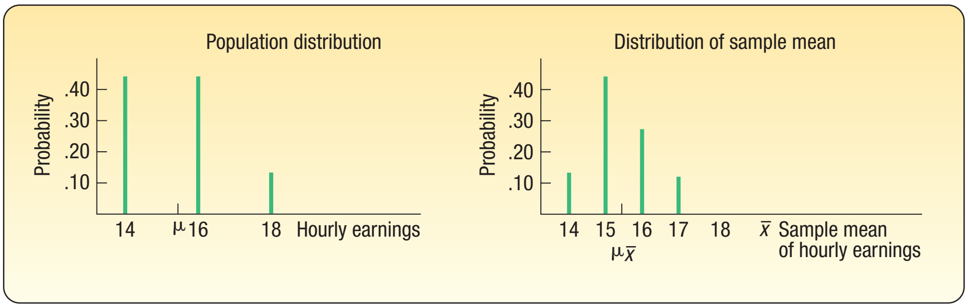


CHART 8–2 Distributions of Population Values and Sample Means

In summary, we took all possible random samples from a population and for each sample calculated a sample statistic (the mean amount earned). This example illustrates important relationships between the population distribution and the sampling distribution of the sample mean:

1. The mean of the sample means is exactly equal to the population mean.
2. The dispersion of the sampling distribution of the sample mean is narrower than the population distribution.
3. The sampling distribution of the sample mean tends to become bell-shaped and to approximate the normal probability distribution.

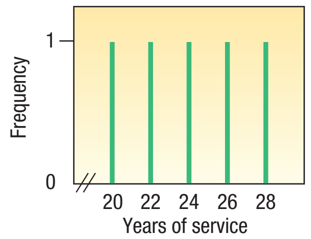
Given a bell-shaped or normal probability distribution, we will be able to apply concepts from Chapter 7 to determine the probability of selecting a sample with a specified sample mean. In the next section, we will show the importance of sample size as it relates to the sampling distribution of the sample mean.

SELF-REVIEW 8–3

The years of service of the five executives employed by Standard Chemicals are:

Name	Years
Mr. Snow	20
Ms. Tolson	22
Mr. Kraft	26
Ms. Irwin	24
Mr. Jones	28

(a) Using the combination formula, how many samples of size 2 are possible?
(b) List all possible samples of two executives from the population and compute their means.
(c) Organize the means into a sampling distribution.
(d) Compare the population mean and the mean of the sample means.
(e) Compare the dispersion in the population with that in the distribution of the sample mean.
(f) A chart portraying the population values follows. Is the distribution of population values normally distributed (bell-shaped)?



(g) Is the distribution of the sample mean computed in part (c) starting to show some tendency toward being bell-shaped?

EXERCISES

5. A population consists of the following four values: 12, 12, 14, and 16.
- **a.** List all samples of size 2, and compute the mean of each sample.
- **b.** Compute the mean of the distribution of the sample mean and the population mean. Compare the two values.
- **c.** Compare the dispersion in the population with that of the sample mean.

6. A population consists of the following five values: 2, 2, 4, 4, and 8.
- **a.** List all samples of size 2, and compute the mean of each sample.
- **b.** Compute the mean of the distribution of sample means and the population mean. Compare the two values.
- **c.** Compare the dispersion in the population with that of the sample means.

7. A population consists of the following five values: 12, 12, 14, 15, and 20.
- **a.** List all samples of size 3, and compute the mean of each sample.
- **b.** Compute the mean of the distribution of sample means and the population mean. Compare the two values.
- **c.** Compare the dispersion in the population with that of the sample means.

8. A population consists of the following five values: 0, 0, 1, 3, 6.
- **a.** List all samples of size 3, and compute the mean of each sample.
- **b.** Compute the mean of the distribution of sample means and the population mean. Compare the two values.
- **c.** Compare the dispersion in the population with that of the sample means.

9. In the law firm Tybo and Associates, there are six partners. Listed is the number of cases each partner actually tried in court last month.

Partner	Number of Cases
Ruud	3
Wu	6
Sass	3
Flores	3
Wilhelms	0
Schueller	1

- **a.** How many different samples of size 3 are possible?
- **b.** List all possible samples of size 3, and compute the mean number of cases in each sample.
- **c.** Compare the mean of the distribution of sample means to the population mean.
- **d.** On a chart similar to Chart 8–2, compare the dispersion in the population with that of the sample means.

10. There are five sales associates at Mid-Motors Ford. The five associates and the number of cars they sold last week are:

Sales Associate	Cars Sold
Peter Hankish	8
Connie Stallter	6
Juan Lopez	4
Ted Barnes	10
Peggy Chu	6

a. How many different samples of size 2 are possible?
b. List all possible samples of size 2, and compute the mean of each sample.
c. Compare the mean of the sampling distribution of the sample mean with that of the population.
d. On a chart similar to Chart 8–2, compare the dispersion in sample means with that of the population.

LO8-4
Recite the central limit theorem and define the mean and standard error of the sampling distribution of the sample mean.

THE CENTRAL LIMIT THEOREM

In this section, we examine the **central limit theorem.** Its application to the sampling distribution of the sample mean, introduced in the previous section, allows us to use the normal probability distribution to create confidence intervals for the population mean (described in Chapter 9) and perform tests of hypothesis (described in Chapter 10). The central limit theorem states that, for large random samples, the shape of the sampling distribution of the sample mean is close to the normal probability distribution. The approximation is more accurate for large samples than for small samples. This is one of the most useful conclusions in statistics. We can reason about the distribution of the sample mean with absolutely no information about the shape of the population distribution from which the sample is taken. In other words, the central limit theorem is true for all population distributions.

CENTRAL LIMIT THEOREM If all samples of a particular size are selected from any population, the sampling distribution of the sample mean is approximately a normal distribution. This approximation improves with larger samples.

To further illustrate the central limit theorem, if the population follows a normal probability distribution, then for any sample size the sampling distribution of the sample mean will also be normal. If the population distribution is symmetrical (but not normal), you will see the normal shape of the distribution of the sample mean emerge with samples as small as 10. On the other hand, if you start with a distribution that is skewed or has thick tails, it may require samples of 30 or more to observe the normality feature. This concept is summarized in Chart 8–3 for various population shapes. Observe the convergence to a normal distribution regardless of the shape of the population distribution.

The idea that a distribution of sample means will converge to normality when the population is not normal is illustrated in Charts 8–4, 8–5, and 8–6. We will discuss this example in more detail shortly, but Chart 8–4 is a graph of a discrete probability distribution that is positively skewed. There are many possible samples of 5 that might be selected from this population. Suppose we randomly select 25 samples of size 5 from the population portrayed in Chart 8–4 and compute the mean of each sample. These results are shown in Chart 8–5. Notice that the shape of the distribution of sample means has changed from the shape of the original population even though we selected only 25 of the many possible samples. To put it another way, we selected 25 random samples of $n = 5$ from a population that is positively skewed and found the distribution

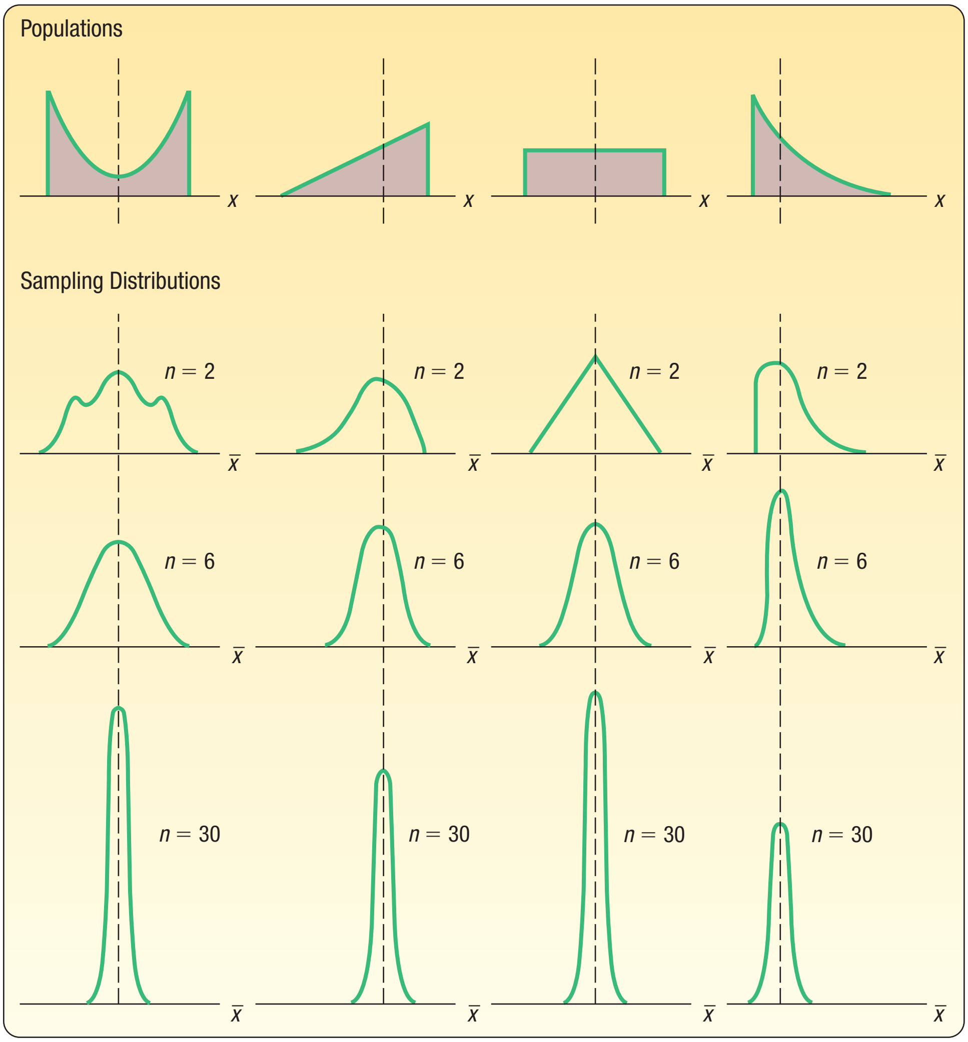


CHART 8–3 Results of the Central Limit Theorem for Several Populations

of sample means is different from the shape of the population. As we take larger samples, that is, $n = 20$ instead of $n = 5$, we will find the distribution of the sample mean will approach the normal distribution. Chart 8–6 shows the results of 25 random samples of 20 observations each from the same population. Observe the clear trend toward the normal probability distribution. This is the point of the central limit theorem. The following example/solution will underscore this condition.

EXAMPLE

Ed Spence began his sprocket business 20 years ago. The business has grown over the years and now employs 40 people. Spence Sprockets Inc. faces some major decisions regarding health care for these employees. Before making a final decision on what health care plan to purchase, Ed decides to form a committee of five representative employees. The committee will be asked to study the health care issue carefully and make a recommendation as to what plan best fits the employees' needs. Ed feels the views of newer employees toward health care may differ from those of more experienced employees. If Ed randomly selects this committee, what can he expect in terms of the mean years with Spence Sprockets for those on the committee? How does the shape of the distribution of years of service of all employees (the population) compare with the shape of the sampling distribution of the mean? The years of service (rounded to the nearest year) of the 40 employees currently on the Spence Sprockets Inc. payroll are as follows.

11	4	18	2	1	2	0	2	2	4
3	4	1	2	2	3	3	19	8	3
7	1	0	2	7	0	4	5	1	14
16	8	9	1	1	2	5	10	2	3

SOLUTION

Chart 8–4 shows a histogram for the frequency distribution of the years of service for the population of 40 current employees. This distribution is positively skewed. Why? Because the business has grown in recent years, the distribution shows that 29 of the 40 employees have been with the company less than 6 years. Also, there are 11 employees who have worked at Spence Sprockets for more than 6 years. In particular, four employees have been with the company 12 years or more (count the frequencies above 12). So there is a long tail in the distribution of service years to the right, that is, the distribution is positively skewed.

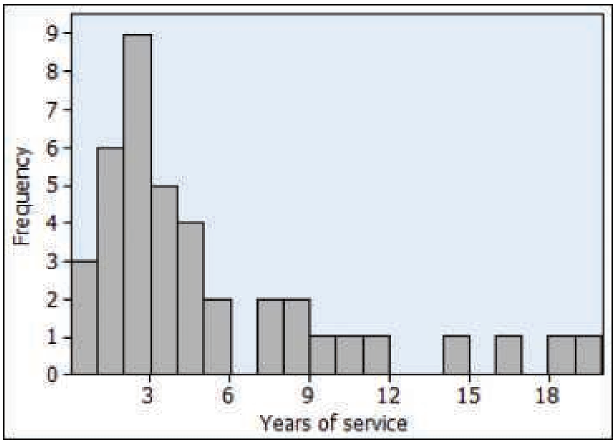


CHART 8–4 Years of Service for Spence Sprockets Inc. Employees

Let's consider the first of Ed Spence's problems. He would like to form a committee of five employees to look into the health care question and suggest what type of health care coverage would be most appropriate for the majority of workers. How should he select the committee? If he selects the committee randomly, what might he expect in terms of mean years of service for those on the committee?

To begin, Ed writes the years of service for each of the 40 employees on pieces of paper and puts them into an old baseball hat. Next, he shuffles the pieces of paper and randomly selects five slips of paper. The years of service for these five employees are 1, 9, 0, 19, and 14 years. Thus, the mean years of service for these five sampled employees is 8.60 years. How does that compare with the population mean? At this point, Ed does not know the population mean, but the number of employees in the population is only 40, so he decides to calculate the mean years of service for *all* his employees. It is 4.8 years, found by adding the years of service for *all* the employees and dividing the total by 40.

$$\mu = \frac{11 + 4 + 18 + \cdots + 2 + 3}{40} = 4.80$$

The difference between a sample mean ($\bar{x}$) and the population mean (μ) is called **sampling error**. In other words, the difference of 3.80 years between the sample mean of 8.60 and the population mean of 4.80 is the sampling error. It is due to chance. Thus, if Ed selected these five employees to constitute the committee, their mean years of service would be larger than the population mean.

What would happen if Ed put the five pieces of paper back into the baseball hat and selected another sample? Would you expect the mean of this second sample to be exactly the same as the previous one? Suppose he selects another sample of five employees and finds the years of service in this sample to be 7, 4, 4, 1, and 3. This sample mean is 3.80 years. The result of selecting 25 samples of five employees and computing the mean for each sample is shown in Table 8–5 and Chart 8–5. There are actually 658,008 possible samples of 5 from the population of 40 employees, found by the combination formula (5–10) for 40 things taken 5 at a time. Notice the difference in the shape of the population and the distribution of these sample means. The population of the years of service for employees (Chart 8–4) is positively skewed, but the distribution of these 25 sample means does not reflect the same positive skew. There is also a difference in the range of the sample means versus the range of the population. The population ranged from 0 to 19 years, whereas the sample means range from 1.6 to 8.6 years.

TABLE 8–5 Twenty-Five Random Samples of Five Employees

	Sample Data						
Sample	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5	Sum	Mean
A	1	9	0	19	14	43	8.6
B	7	4	4	1	3	19	3.8
C	8	19	8	2	1	38	7.6
D	4	18	2	0	11	35	7.0
E	4	2	4	7	18	35	7.0
F	1	2	0	3	2	8	1.6
G	2	3	2	0	2	9	1.8
H	11	2	9	2	4	28	5.6
I	9	0	4	2	7	22	4.4
J	1	1	1	11	1	15	3.0
K	2	0	0	10	2	14	2.8
L	0	2	3	2	16	23	4.6
M	2	3	1	1	1	8	1.6
N	3	7	3	4	3	20	4.0
O	1	2	3	1	4	11	2.2
P	19	0	1	3	8	31	6.2
Q	5	1	7	14	9	36	7.2
R	5	4	2	3	4	18	3.6
S	14	5	2	2	5	28	5.6
T	2	1	1	4	7	15	3.0
U	3	7	1	2	1	14	2.8
V	0	1	5	1	2	9	1.8
W	0	3	19	4	2	28	5.6
X	4	2	3	4	0	13	2.6
Y	1	1	2	3	2	9	1.8

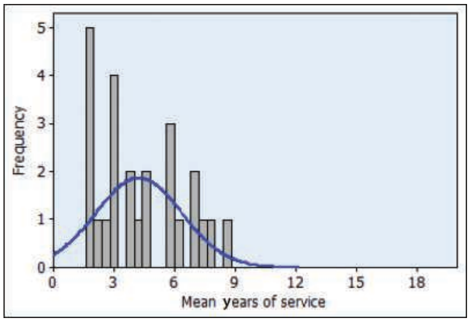


CHART 8–5 Histogram of Mean Years of Service for 25 Samples of Five Employees

Now let's change the example by increasing the size of each sample from 5 employees to 20. Table 8–6 reports the result of selecting 25 samples of 20 employees each and computing their sample means. These sample means are shown graphically in Chart 8–6. Compare the shape of this distribution to the population (Chart 8–4) and to the distribution of sample means where the sample is $n = 5$ (Chart 8–5). You should observe two important features:

1. The shape of the distribution of the sample mean is different from that of the population. In Chart 8–4, the distribution of all employees is positively skewed. However, as we select random samples from this population, the shape of the

TABLE 8–6 Twenty-Five Random Samples of 20 Employees

	Sample Data							
Sample	Obs 1	Obs 2	Obs 3	–	Obs 19	Obs 20	Sum	Mean
A	3	8	3	–	4	16	79	3.95
B	2	3	8	–	3	1	65	3.25
C	14	5	0	–	19	8	119	5.95
D	9	2	1	–	1	3	87	4.35
E	18	1	2	–	3	14	107	5.35
F	10	4	4	–	2	1	80	4.00
G	5	7	11	–	2	4	131	6.55
H	3	0	2	–	16	5	85	4.25
I	0	0	18	–	2	3	80	4.00
J	2	7	2	–	3	2	81	4.05
K	7	4	5	–	1	2	84	4.20
L	0	3	10	–	0	4	81	4.05
M	4	1	2	–	1	2	88	4.40
N	3	16	1	–	11	1	95	4.75
O	2	19	2	–	2	2	102	5.10
P	2	18	16	–	4	3	100	5.00
Q	3	2	3	–	3	1	102	5.10
R	2	3	1	–	0	2	73	3.65
S	2	14	19	–	0	7	142	7.10
T	0	1	3	–	2	0	61	3.05
U	1	0	1	–	9	3	65	3.25
V	1	9	4	–	2	11	137	6.85
W	8	1	9	–	8	7	107	5.35
X	4	2	0	–	2	5	86	4.30
Y	1	2	1	–	1	18	101	5.05

distribution of the sample mean changes. As we increase the size of the sample, the distribution of the sample mean approaches the normal probability distribution. This illustrates the central limit theorem.

2. There is less dispersion in the sampling distribution of the sample mean than in the population distribution. In the population, the years of service ranged from 0 to 19 years. When we selected samples of 5, the sample means ranged from 1.6 to 8.6 years, and when we selected samples of 20, the means ranged from 3.05 to 7.10 years.

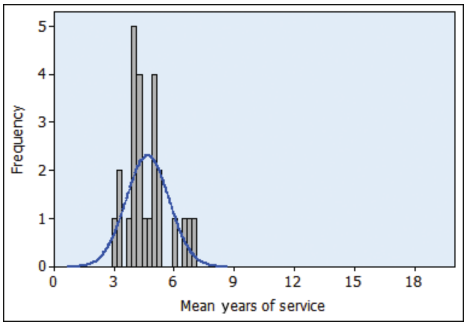


CHART 8–6 Histogram of Mean Years of Service for 25 Samples of 20 Employees

We can also compare the mean of the sample means to the population mean. The mean of the 25 samples of 20 employees reported in Table 8–6 is 4.676 years.

$$\mu_{\bar{x}} = \frac{3.95 + 3.25 + \cdots + 4.30 + 5.05}{25} = 4.676$$

We use the symbol $\mu_{\bar{x}}$ to identify the mean of the distribution of the sample mean. The subscript reminds us that the distribution is of the sample mean. It is read "mu sub x bar." We observe that the mean of the sample means, 4.676 years, is very close to the population mean of 4.80.

What should we conclude from this example? The central limit theorem indicates that, regardless of the shape of the population distribution, the sampling distribution of the sample mean will move toward the normal probability distribution. The larger the number of observations sampled or selected, the stronger the convergence. The Spence Sprockets Inc. example shows how the central limit theorem works. We began with a positively skewed population (Chart 8–4). Next, we selected 25 random samples of 5 observations, computed the mean of each sample, and finally organized these 25 sample means into a histogram (Chart 8–5). We observe that the shape of the sampling distribution of the sample mean is very different from that of the population. The population distribution is positively skewed compared to the nearly normal shape of the sampling distribution of the sample mean.

To further illustrate the effects of the central limit theorem, we increased the number of observations in each sample from 5 to 20. We selected 25 samples of 20 observations each and calculated the mean of each sample. Finally, we organized these sample means into a histogram (Chart 8–6). The shape of the histogram in Chart 8–6 is clearly moving toward the normal probability distribution.

If you go back to Chapter 6 where several binomial distributions with a "success" proportion of .10 are shown in Chart 6–3 on page 190, you can see yet another demonstration of the central limit theorem. Observe as n increases from 7 through 12 and 20 up to 40 that the profile of the probability distributions moves closer and closer to a normal probability distribution. Chart 8–6 also shows the convergence to normality as n increases. This again reinforces the fact that, as more observations are sampled from any population distribution, the shape of the sampling distribution of the sample mean will get closer and closer to a normal distribution.

The **central limit theorem,** defined on page 265, does not say anything about the dispersion of the sampling distribution of the sample mean or about the comparison of the mean of the sampling distribution of the sample mean to the mean of the population. However, in our Spence Sprockets example, we did observe that there was less dispersion in the distribution of the sample mean than in the population distribution by noting the difference in the range in the population and the range of the sample means. We observe that the mean of the sample means is close to the mean of the population. It can be demonstrated that the mean of the sampling distribution is exactly equal to the population mean (i.e., $\mu_{\bar{x}} = \mu$), and if the standard deviation in the population is σ, the standard deviation of the sample means is $\sigma/\sqrt{n}$ where n is the number of observations in each sample. We refer to $\sigma/\sqrt{n}$ as the **standard error of the mean.** Its longer name is actually the *standard deviation of the sampling distribution of the sample mean*.

STANDARD ERROR OF THE MEAN

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \qquad \textbf{(8–1)}$$

In this section, we also came to other important conclusions.

1. The mean of the distribution of sample means will be *exactly* equal to the population mean if we are able to select all possible samples of the same size from a given population. That is:

$$\mu = \mu_{\bar{x}}$$

   Even if we do not select all samples, we can expect the mean of the distribution of sample means to be close to the population mean.
2. There will be less dispersion in the sampling distribution of the sample mean than in the population. If the standard deviation of the population is σ, the standard deviation of the distribution of sample means is $\sigma/\sqrt{n}$. Note that when we increase the size of the sample, the standard error of the mean decreases.

SELF-REVIEW 8–4

Refer to the Spence Sprockets Inc. data on page 267. Select 10 random samples of five employees each. Use the methods described earlier in the chapter and the Table of Random Numbers (Appendix B.4) to find the employees to include in the sample. Compute the mean of each sample and plot the sample means on a chart similar to Chart 8–4. What is the mean of your 10 sample means?

EXERCISES

11. FILE Appendix B.4 is a table of random numbers that are uniformly distributed. Hence, each digit from 0 to 9 has the same likelihood of occurrence.

a. Draw a graph showing the population distribution of random numbers. What is the population mean?

b. Following are the first 10 rows of five digits from the table of random numbers in Appendix B.4. Assume that these are 10 random samples of five values each. Determine the mean of each sample and plot the means on a chart similar to

Chart 8–4. Compare the mean of the sampling distribution of the sample mean with the population mean.

0	2	7	1	1
9	4	8	7	3
5	4	9	2	1
7	7	6	4	0
6	1	5	4	5
1	7	1	4	7
1	3	7	4	8
8	7	4	5	5
0	8	9	9	9
7	8	8	0	4

12. FILE Scrapper Elevator Company has 20 sales representatives who sell its product throughout the United States and Canada. The number of units sold last month by each representative is listed below. Assume these sales figures to be the population values.

2	3	2	3	3	4	2	4	3	2	2	7	3	4	5	3	3	3	3	5

a. Draw a graph showing the population distribution.
b. Compute the mean of the population.
c. Select five random samples of 5 each. Compute the mean of each sample. Use the methods described in this chapter and Appendix B.4 to determine the items to be included in the sample.
d. Compare the mean of the sampling distribution of the sample mean to the population mean. Would you expect the two values to be about the same?
e. Draw a histogram of the sample means. Do you notice a difference in the shape of the distribution of sample means compared to the shape of the population distribution?

13. Consider all of the coins (pennies, nickels, quarters, etc.) in your pocket or purse as a population. Make a frequency table beginning with the current year and counting backward to record the ages (in years) of the coins. For example, if the current year is 2017, then a coin with 2015 stamped on it is 2 years old.
a. Draw a histogram or other graph showing the population distribution.
b. Randomly select five coins and record the mean age of the sampled coins. Repeat this sampling process 20 times. Now draw a histogram or other graph showing the distribution of the sample means.
c. Compare the shapes of the two histograms.

14. Consider the digits in the phone numbers on a randomly selected page of your local phone book a population. Make a frequency table of the final digit of 30 randomly selected phone numbers. For example, if a phone number is 555-9704, record a 4.
a. Draw a histogram or other graph of this population distribution. Using the uniform distribution, compute the population mean and the population standard deviation.
b. Also record the sample mean of the final four digits (9704 would lead to a mean of 5). Now draw a histogram or other graph showing the distribution of the sample means.
c. Compare the shapes of the two histograms.

LO8-5
Apply the central limit theorem to calculate probabilities.

USING THE SAMPLING DISTRIBUTION OF THE SAMPLE MEAN

The previous discussion is important because most business decisions are made on the basis of sample information. Here are some examples.

1. Arm & Hammer Company wants to ensure that its laundry detergent actually contains 100 fluid ounces, as indicated on the label. Historical summaries from the filling process indicate the mean amount per container is 100 fluid ounces and the standard deviation is 2 fluid ounces. At 10 a.m., a quality technician measures 40 containers and finds the mean amount per container is 99.8 fluid ounces. Should the technician shut down the filling operation?
2. A. C. Nielsen Company provides information to organizations advertising on television. Prior research indicates that adult Americans watch an average of 6.0 hours per day of television. The standard deviation is 1.5 hours. What is the probability that we could randomly select a sample of 50 adults and find that they watch an average of 6.5 hours or more of television per day?
3. Haughton Elevator Company wishes to develop specifications for the number of people who can ride in a new oversized elevator. Suppose the mean weight of an adult is 160 pounds and the standard deviation is 15 pounds. However, the distribution of weights does not follow the normal probability distribution. It is positively skewed. For a sample of 30 adults, what is the likelihood that their mean weight is 170 pounds or more?

We can answer the questions in each of these situations using the ideas discussed in the previous section. In each case, we have a population with information about its mean and standard deviation. Using this information and sample size, we can determine the distribution of sample means and compute the probability that a sample mean will fall within a certain range. The sampling distribution will be normally distributed under two conditions:

1. When the samples are taken from populations known to follow the normal distribution. In this case, the size of the sample is not a factor.
2. When the shape of the population distribution is not known, sample size is important. In general, the sampling distribution will be normally distributed as the sample size approaches infinity. In practice, a sampling distribution will be close to a normal distribution with samples of at least 30 observations.

We use formula (7–5) from the previous chapter to convert any normal distribution to the standard normal distribution. Using formula (7–5) to compute z values, we can use the standard normal table, Appendix B.3, to find the probability that an observation is within a specific range. The formula for finding a z value is:

$$z = \frac{x - \mu}{\sigma}$$

In this formula, x is the value of the random variable, μ is the population mean, and σ is the population standard deviation.

However, when we sample from populations, we are interested in the distribution of $\overline{X}$, the sample mean, instead of X, the value of one observation. That is the first change we make in formula (7–5). The second is that we use the standard error of the mean of n observations instead of the population standard deviation. That is, we use $\sigma/\sqrt{n}$ in the denominator rather than σ. Therefore, to find the likelihood of a sample mean within a specified range, we first use the following formula to find the corresponding z value. Then we use Appendix B.3 or statistical software to determine the probability.

FINDING THE z VALUE OF $\bar{x}$ WHEN THE POPULATION STANDARD DEVIATION IS KNOWN

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \qquad \textbf{(8–2)}$$

The following example/solution will show the application.

EXAMPLE

The Quality Assurance Department for Cola, Inc. maintains records regarding the amount of cola in its jumbo bottle. The actual amount of cola in each bottle is critical but varies a small amount from one bottle to the next. Cola, Inc. does not wish to underfill the bottles because it will have a problem with truth in labeling. On the other hand, it cannot overfill each bottle because it would be giving cola away, hence reducing its profits. Records maintained by the Quality Assurance Department indicate that the amount of cola follows the normal probability distribution. The mean amount per bottle is 31.2 ounces and the population standard deviation is 0.4 ounce. At 8 a.m. today the quality technician randomly selected 16 bottles from the filling line. The mean amount of cola contained in the bottles is 31.38 ounces. Is this an unlikely result? Is it likely the process is putting too much soda in the bottles? To put it another way, is the sampling error of 0.18 ounce unusual?

SOLUTION

We use the results of the previous section to find the likelihood that we could select a sample of 16 (n) bottles from a normal population with a mean of 31.2 (μ) ounces and a population standard deviation of 0.4 (σ) ounce and find the sample mean to be 31.38 ($\bar{x}$) or more. We use formula (8–2) to find the value of z.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{31.38 - 31.20}{0.4/\sqrt{16}} = 1.80$$

The numerator of this equation, $\bar{x} - \mu = 31.38 - 31.20 = .18$, is the sampling error. The denominator, $\sigma/\sqrt{n} = 0.4/\sqrt{16} = 0.1$, is the standard error of the sampling distribution of the sample mean. So the z values express the sampling error in standard units—in other words, the standard error.

Next, we compute the likelihood of a z value greater than 1.80. In Appendix B.3, locate the probability corresponding to a z value of 1.80. It is .4641. The likelihood of a z value greater than 1.80 is .0359, found by .5000 − .4641.

What do we conclude? It is unlikely, less than a 4% chance, we could select a sample of 16 observations from a normal population with a mean of 31.2 ounces and a population standard deviation of 0.4 ounce and find the sample mean equal to or greater than 31.38 ounces. We conclude the process is putting too much cola in the bottles. The quality technician should see the production supervisor about reducing the amount of soda in each bottle. This information is summarized in Chart 8–7.

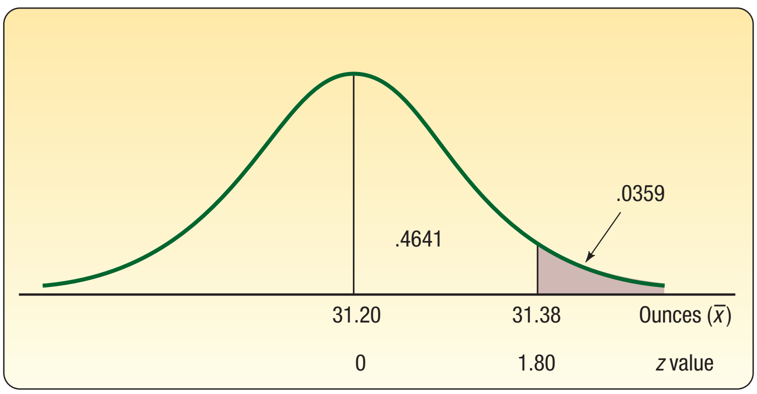


CHART 8–7 Sampling Distribution of the Mean Amount of Cola in a Jumbo Bottle

SELF-REVIEW 8–5

Refer to the Cola, Inc. information. Suppose the quality technician selected a sample of 16 jumbo bottles that averaged 31.08 ounces. What can you conclude about the filling process?

EXERCISES

15. A normal population has a mean of 60 and a standard deviation of 12. You select a random sample of 9. Compute the probability the sample mean is:
- **a.** Greater than 63.
- **b.** Less than 56.
- **c.** Between 56 and 63.

16. A normal population has a mean of 75 and a standard deviation of 5. You select a sample of 40. Compute the probability the sample mean is:
- **a.** Less than 74.
- **b.** Between 74 and 76.
- **c.** Between 76 and 77.
- **d.** Greater than 77.

17. In a certain section of Southern California, the distribution of monthly rent for a one-bedroom apartment has a mean of $2,200 and a standard deviation of $250. The distribution of the monthly rent does not follow the normal distribution. In fact, it is positively skewed. What is the probability of selecting a sample of 50 one-bedroom apartments and finding the mean to be at least $1,950 per month?

18. According to an IRS study, it takes a mean of 330 minutes for taxpayers to prepare, copy, and electronically file a 1040 tax form. This distribution of times follows the normal distribution and the standard deviation is 80 minutes. A consumer watchdog agency selects a random sample of 40 taxpayers.
- **a.** What is the standard error of the mean in this example?
- **b.** What is the likelihood the sample mean is greater than 320 minutes?
- **c.** What is the likelihood the sample mean is between 320 and 350 minutes?
- **d.** What is the likelihood the sample mean is greater than 350 minutes?

CHAPTER SUMMARY

I. The characteristics of the *F* distribution are:
- **A.** There are many reasons for sampling a population.
- **B.** The results of a sample may adequately estimate the value of the population parameter, thus saving time and money.
- **C.** It may be too time-consuming to contact all members of the population.
- **D.** It may be impossible to check or locate all the members of the population.
- **E.** The cost of studying all the items in the population may be prohibitive.
- **F.** Often testing destroys the sampled item and it cannot be returned to the population.

II. In an unbiased or probability sample, all members of the population have a chance of being selected for the sample. There are several probability sampling methods.
- **A.** In a simple random sample, all members of the population have the same chance of being selected for the sample.
- **B.** In a systematic sample, a random starting point is selected, and then every *k*th item thereafter is selected for the sample.
- **C.** In a stratified sample, the population is divided into several groups, called strata, and then a random sample is selected from each stratum.
- **D.** In cluster sampling, the population is divided into primary units, then samples are drawn from the primary units.

III. The sampling error is the difference between a population parameter and a sample statistic.

IV. The sampling distribution of the sample mean is a probability distribution of all possible sample means of the same sample size.

A. For a given sample size, the mean of all possible sample means selected from a population is equal to the population mean.

B. There is less variation in the distribution of the sample mean than in the population distribution.

C. The standard error of the mean measures the variation in the sampling distribution of the sample mean. The standard error is found by:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \tag{8–1}$$

D. If the population follows a normal distribution, the sampling distribution of the sample mean will also follow the normal distribution for samples of any size. If the population is not normally distributed, the sampling distribution of the sample mean will approach a normal distribution when the sample size is at least 30. Assume the population standard deviation is known. To determine the probability that a sample mean falls in a particular region, use the following formula.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \tag{8–2}$$

PRONUNCIATION KEY

SYMBOL	MEANING	PRONUNCIATION
$\mu_{\bar{x}}$	Mean of the sampling distribution of the sample mean	*mu sub x bar*
$\sigma_{\bar{x}}$	Population standard error of the sample mean	*sigma sub x bar*

CHAPTER EXERCISES

19. The 25 retail stores located in the North Towne Square Mall numbered 00 through 24 are:

00	Elder-Beerman	09	Lion Store	18	County Seat
01	Sears	10	Bootleggers	19	Kid Mart
02	Deb Shop	11	Formal Man	20	Lerner
03	Frederick's of Hollywood	12	Leather Ltd.	21	Coach House Gifts
04	Petries	13	Barnes and Noble	22	Spencer Gifts
05	Easy Dreams	14	Pat's Hallmark	23	CPI Photo Finish
06	Summit Stationers	15	Things Remembered	24	Regis Hairstylists
07	E. B. Brown Opticians	16	Pearle Vision Express		
08	Kay-Bee Toy & Hobby	17	Dollar Tree		

a. If the following random numbers are selected, which retail stores should be contacted for a survey? 11, 65, 86, 62, 06, 10, 12, 77, and 04

b. Select a random sample of four retail stores. Use Appendix B.4.

c. A systematic sampling procedure will be used. The first store will be selected and then every third store. Which stores will be in the sample?

20. The Medical Assurance Company is investigating the cost of a routine office visit to family-practice physicians in the Rochester, New York, area. The following is a list of 39 family-practice physicians in the region. Physicians are to be randomly selected and contacted regarding their charges. The 39 physicians have been coded from 00 to 38. Also noted is whether they are in practice by themselves (S), have a partner (P), or are in a group practice (G).

Number	Physician	Type of Practice	Number	Physician	Type of Practice
00	R. E. Scherbarth, M.D.	S	20	Gregory Yost, M.D.	P
01	Crystal R. Goveia, M.D.	P	21	J. Christian Zona, M.D.	P
02	Mark D. Hillard, M.D.	P	22	Larry Johnson, M.D.	P
03	Jeanine S. Huttner, M.D.	P	23	Sanford Kimmel, M.D.	P
04	Francis Aona, M.D.	P	24	Harry Mayhew, M.D.	S
05	Janet Arrowsmith, M.D.	P	25	Leroy Rodgers, M.D.	S
06	David DeFrance, M.D.	S	26	Thomas Tafelski, M.D.	S
07	Judith Furlong, M.D.	S	27	Mark Zilkoski, M.D.	G
08	Leslie Jackson, M.D.	G	28	Ken Bertka, M.D.	G
09	Paul Langenkamp, M.D.	S	29	Mark DeMichiei, M.D.	G
10	Philip Lepkowski, M.D.	S	30	John Eggert, M.D.	P
11	Wendy Martin, M.D.	S	31	Jeanne Fiorito, M.D.	P
12	Denny Mauricio, M.D.	P	32	Michael Fitzpatrick, M.D.	P
13	Hasmukh Parmar, M.D.	P	33	Charles Holt, D.O.	P
14	Ricardo Pena, M.D.	P	34	Richard Koby, M.D.	P
15	David Reames, M.D.	P	35	John Meier, M.D.	P
16	Ronald Reynolds, M.D.	G	36	Douglas Smucker, M.D.	S
17	Mark Steinmetz, M.D.	G	37	David Weldy, M.D.	P
18	Geza Torok, M.D.	S	38	Cheryl Zaborowski, M.D.	P
19	Mark Young, M.D.	P			

a. The random numbers obtained from Appendix B.4 are 31, 94, 43, 36, 03, 24, 17, and 09. Which physicians should be contacted?

b. Select a random sample of four physicians using the random numbers of Appendix B.4.

c. Using systematic random sampling, every fifth physician is selected starting with the fourth physician in the list. Which physicians will be contacted?

d. Select a sample that includes two physicians in solo practice (S), two in partnership (P), and one in group practice (G). Explain your procedure.

21. A population consists of the following three values: 1, 2, and 3.

a. Sampling with replacement, list all possible samples of size 2 and compute the mean of every sample.

b. Find the means of the distribution of the sample mean and the population mean. Compare the two values.

c. Compare the dispersion of the population with that of the sample mean.

d. Describe the shapes of the two distributions.

22. Based on all student records at Camford University, students spend an average of 5.5 hours per week playing organized sports. The population's standard deviation is 2.2 hours per week. Based on a sample of 121 students, Healthy Lifestyles Incorporated (HLI) would like to apply the central limit theorem to make various estimates.

a. Compute the standard error of the sample mean.

b. What is the chance HLI will find a sample mean between 5 and 6 hours?

c. Calculate the probability that the sample mean will be between 5.3 and 5.7 hours.

d. How strange would it be to obtain a sample mean greater than 6.5 hours?

23. The manufacturer of eComputers, an economy-priced computer, recently completed the design for a new laptop model. eComputer's top management would like some assistance in pricing the new laptop. Two market research firms were contacted and asked to prepare a pricing strategy. Marketing-Gets-Results tested the new eComputers laptop with 50 randomly selected consumers who indicated they plan to purchase a laptop within the next year. The second marketing research firm, called Marketing-Reaps-Profits, test-marketed the new eComputers laptop with 200 current laptop owners. Which of the marketing research companies' test results will be more useful? Discuss why.

24. Answer the following questions in one or two well-constructed sentences.
  a. What happens to the standard error of the mean if the sample size is increased?
  b. What happens to the distribution of the sample means if the sample size is increased?
  c. When using sample means to estimate the population mean, what is the benefit of using larger sample sizes?

25. There are 25 motels in Goshen, Indiana. The number of rooms in each motel follows:

90	72	75	60	75	72	84	72	88	74	105	115	68	74	80	64	104	82	48	58	60	80	48	58	100

  a. Using a table of random numbers (Appendix B.4), select a random sample of five motels from this population.
  b. Obtain a systematic sample by selecting a random starting point among the first five motels and then select every fifth motel.
  c. Suppose the last five motels are "cut-rate" motels. Describe how you would select a random sample of three regular motels and two cut-rate motels.

26. As a part of their customer-service program, United Airlines randomly selected 10 passengers from today's 9 a.m. Chicago–Tampa flight. Each sampled passenger will be interviewed about airport facilities, service, and so on. To select the sample, each passenger was given a number on boarding the aircraft. The numbers started with 001 and ended with 250.
  a. Select 10 usable numbers at random using Appendix B.4.
  b. The sample of 10 could have been chosen using a systematic sample. Choose the first number using Appendix B.4, and then list the numbers to be interviewed.
  c. Evaluate the two methods by giving the advantages and possible disadvantages.
  d. What other way could a random sample be selected from the 250 passengers?

27. Suppose your statistics instructor gave six examinations during the semester. You received the following exam scores (percent correct): 79, 64, 84, 82, 92, and 77. To compute your final course grade, the instructor decided to randomly select two exam scores, compute their mean, and use this score to determine your final course grade.
  a. Compute the population mean.
  b. How many different samples of two test grades are possible?
  c. List all possible samples of size 2 and compute the mean of each.
  d. Compute the mean of the sample means and compare it to the population mean.
  e. If you were a student, would you like this arrangement? Would the result be different from dropping the lowest score? Write a brief report.

28. At the downtown office of First National Bank, there are five tellers. Last week, the tellers made the following number of errors each: 2, 3, 5, 3, and 5.
  a. How many different samples of two tellers are possible?
  b. List all possible samples of size 2 and compute the mean of each.
  c. Compute the mean of the sample means and compare it to the population mean.

29. The Quality Control Department employs five technicians during the day shift. Listed below is the number of times each technician instructed the production foreman to shut down the manufacturing process last week.

Technician	Shutdowns	Technician	Shutdowns
Taylor	4	Rousche	3
Hurley	3	Huang	2
Gupta	5		

  a. How many different samples of two technicians are possible from this population?
  b. List all possible samples of two observations each and compute the mean of each sample.
  c. Compare the mean of the sample means with the population mean.
  d. Compare the shape of the population distribution with the shape of the distribution of the sample means.

30. The Appliance Center has six sales representatives at its North Jacksonville outlet. The following table lists the number of refrigerators sold by each representative last month.

Sales Representative	Number Sold	Sales Representative	Number Sold
Zina Craft	54	Jan Niles	48
Woon Junge	50	Molly Camp	50
Ernie DeBrul	52	Rachel Myak	52

a. How many samples of size 2 are possible?
b. Select all possible samples of size 2 and compute the mean number sold.
c. Organize the sample means into a frequency distribution.
d. What is the mean of the population? What is the mean of the sample means?
e. What is the shape of the population distribution?
f. What is the shape of the distribution of the sample mean?

31. Power +, Inc. produces AA batteries used in remote-controlled toy cars. The mean life of these batteries follows the normal probability distribution with a mean of 35.0 hours and a standard deviation of 5.5 hours. As a part of its quality assurance program, Power +, Inc. tests samples of 25 batteries.
a. What can you say about the shape of the distribution of the sample mean?
b. What is the standard error of the distribution of the sample mean?
c. What proportion of the samples will have a mean useful life of more than 36 hours?
d. What proportion of the samples will have a mean useful life greater than 34.5 hours?
e. What proportion of the samples will have a mean useful life between 34.5 and 36.0 hours?

32. Majesty Video Production Inc. wants the mean length of its advertisements to be 30 seconds. Assume the distribution of ad length follows the normal distribution with a population standard deviation of 2 seconds. Suppose we select a sample of 16 ads produced by Majesty.
a. What can we say about the shape of the distribution of the sample mean time?
b. What is the standard error of the mean time?
c. What percent of the sample means will be greater than 31.25 seconds?
d. What percent of the sample means will be greater than 28.25 seconds?
e. What percent of the sample means will be greater than 28.25 but less than 31.25 seconds?

33. Recent studies indicate that the typical 50-year-old woman spends $350 per year for personal-care products. The distribution of the amounts spent follows a normal distribution with a standard deviation of $45 per year. We select a random sample of 40 women. The mean amount spent for those sampled is $335. What is the likelihood of finding a sample mean this large or larger from the specified population?

34. Information from the American Institute of Insurance indicates the mean amount of life insurance per household in the United States is $165,000. This distribution follows the normal distribution with a standard deviation of $40,000.
a. If we select a random sample of 50 households, what is the standard error of the mean?
b. What is the expected shape of the distribution of the sample mean?
c. What is the likelihood of selecting a sample with a mean of at least $167,000?
d. What is the likelihood of selecting a sample with a mean of more than $155,000?
e. Find the likelihood of selecting a sample with a mean of more than $155,000 but less than $167,000.

35. In the United States, the mean age of men when they marry for the first time follows the normal distribution with a mean of 29 years. The standard deviation of the distribution is 2.5 years. For a random sample of 60 men, what is the likelihood that the age when they were first married is less than 29.3. years?

36. A recent study by the Greater Los Angeles Taxi Drivers Association showed that the mean fare charged for service from Hermosa Beach to Los Angeles International Airport is $21 and the standard deviation is $3.50. We select a sample of 15 fares.
a. What is the likelihood that the sample mean is between $20 and $23?
b. What must you assume to make the above calculation?

37. Crossett Trucking Company claims that the mean weight of its delivery trucks when they are fully loaded is 6,000 pounds and the standard deviation is 150 pounds. Assume that the population follows the normal distribution. Forty trucks are randomly selected and weighed. Within what limits will 95% of the sample means occur?

38. The mean amount purchased by a typical customer at Churchill's Grocery Store is $23.50, with a standard deviation of $5.00. Assume the distribution of amounts purchased follows the normal distribution. For a sample of 50 customers, answer the following questions.

a. What is the likelihood the sample mean is at least $25.00?
b. What is the likelihood the sample mean is greater than $22.50 but less than $25.00?
c. Within what limits will 90% of the sample means occur?

39. The mean performance score on a physical fitness test for Division I student-athletes is 947 with a standard deviation of 205. If you select a random sample of 60 of these students, what is the probability the mean is below 900?

40. Suppose we roll a fair die two times.

a. How many different samples are there?
b. List each of the possible samples and compute the mean.
c. On a chart similar to Chart 8–2, compare the distribution of sample means with the distribution of the population.
d. Compute the mean and the standard deviation of each distribution and compare them.

41. **FILE** Following is a list of the 50 states with the numbers 0 through 49 assigned to them.

Number	State	Number	State
0	Alabama	25	Montana
1	Alaska	26	Nebraska
2	Arizona	27	Nevada
3	Arkansas	28	New Hampshire
4	California	29	New Jersey
5	Colorado	30	New Mexico
6	Connecticut	31	New York
7	Delaware	32	North Carolina
8	Florida	33	North Dakota
9	Georgia	34	Ohio
10	Hawaii	35	Oklahoma
11	Idaho	36	Oregon
12	Illinois	37	Pennsylvania
13	Indiana	38	Rhode Island
14	Iowa	39	South Carolina
15	Kansas	40	South Dakota
16	Kentucky	41	Tennessee
17	Louisiana	42	Texas
18	Maine	43	Utah
19	Maryland	44	Vermont
20	Massachusetts	45	Virginia
21	Michigan	46	Washington
22	Minnesota	47	West Virginia
23	Mississippi	48	Wisconsin
24	Missouri	49	Wyoming

a. You wish to select a sample of eight from this list. The selected random numbers are 45, 15, 81, 09, 39, 43, 90, 26, 06, 45, 01, and 42. Which states are included in the sample?
b. Select a systematic sample of every sixth item using the digit 02 as the starting point. Which states are included?

42. Human Resource Consulting (HRC) surveyed a random sample of 60 Twin Cities construction companies to find information on the costs of their health care plans. One of the items being tracked is the annual deductible that employees must pay. The Minnesota Department of Labor reports that historically the mean deductible amount per employee is $502 with a standard deviation of $100.

a. Compute the standard error of the sample mean for HRC.
b. What is the chance HRC finds a sample mean between \$477 and \$527?
c. Calculate the likelihood that the sample mean is between \$492 and \$512.
d. What is the probability the sample mean is greater than \$550?

43. Over the past decade, the mean number of hacking attacks experienced by members of the Information Systems Security Association is 510 per year with a standard deviation of 14.28 attacks. The number of attacks per year is normally distributed. Suppose nothing in this environment changes.
a. What is the likelihood this group will suffer an average of more than 600 attacks in the next 10 years?
b. Compute the probability the mean number of attacks over the next 10 years is between 500 and 600.
c. What is the possibility they will experience an average of less than 500 attacks over the next 10 years?

44. An economist uses the price of a gallon of milk as a measure of inflation. She finds that the average price is \$3.82 per gallon and the population standard deviation is \$0.33. You decide to sample 40 convenience stores, collect their prices for a gallon of milk, and compute the mean price for the sample.
a. What is the standard error of the mean in this experiment?
b. What is the probability that the sample mean is between \$3.78 and \$3.86?
c. What is the probability that the difference between the sample mean and the population mean is less than \$0.01?
d. What is the likelihood the sample mean is greater than \$3.92?

45. Nike's annual report says that the average American buys 6.5 pairs of sports shoes per year. Suppose a sample of 81 customers is surveyed and the population standard deviation of sports shoes purchased per year is 2.1.
a. What is the standard error of the mean in this experiment?
b. What is the probability that the sample mean is between 6 and 7 pairs of sports shoes?
c. What is the probability that the difference between the sample mean and the population mean is less than 0.25 pair?
d. What is the likelihood the sample mean is greater than 7 pairs?

DATA ANALYTICS

46. **FILE** Refer to the North Valley Real Estate data, which report information on the homes sold last year. Assume the 105 homes is a population. Compute the population mean and the standard deviation of price. Select a sample of 10 homes. Compute the mean. Determine the likelihood of a sample mean price this high or higher.

47. **FILE** Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season. Over the last decade, the mean attendance per team followed a normal distribution with a mean of 2.45 million per team and a standard deviation of .71 million. Compute the mean attendance per team for the 2016 season. Determine the likelihood of a sample mean attendance this large or larger from the population.

48. **FILE** Refer to the Lincolnville School District bus data. Information provided by manufacturers of school buses suggests the mean maintenance cost per year is \$4,400 per bus with a standard deviation of \$1,000. Compute the mean maintenance cost for the Lincolnville buses. Does the Lincolnville data seem to be in line with that reported by the manufacturer? Specifically, what is the probability of Lincolnville's mean annual maintenance cost, or greater, given the manufacturer's data?